

Joint optimization of rate control, routing, and link scheduling

Controlling 3 knobs:

- Sending rates of hosts
- Link scheduling weights
- Traffic split across multiple paths

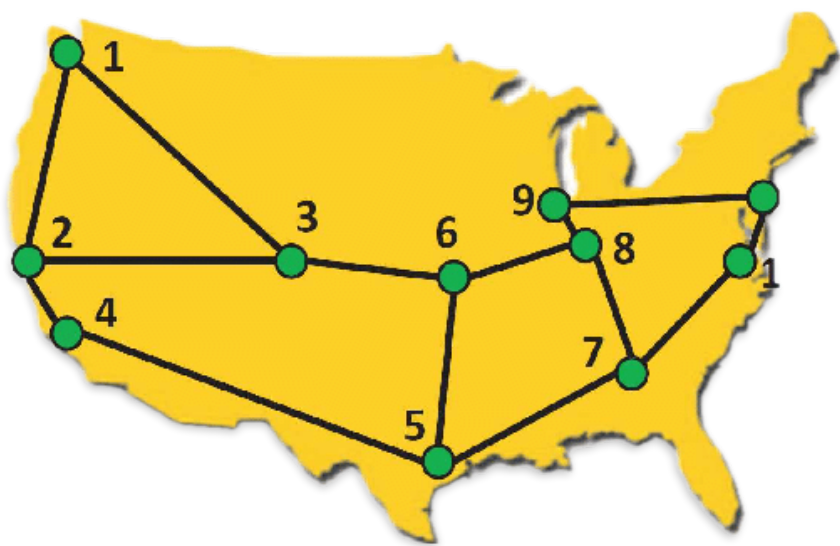


Fig 1. Abilene network topology

Fig 2. A semi-centralized 3-tier design with three types of traffic management entities:

1. a single **Link Coordinator (LC)** on tier 1
 2. multiple **Class Allocators (CA)** on tier 2, one per class
 3. multiple **Datacenter Allocators (DA)** on tier 3, one per class per DC
- LC optimizes aggregate link bandwidth across classes
 - CA optimizes aggregate link bandwidth across datacenters sending traffic in their own classes
 - DA optimizes sending rates across network flows in its own class and originating from its own datacenter

